Al-Balqa Applied University
Faculty of Engineering Technology
Department of Scientific Basic Sciences
Calculus 1 Mid Exam

Name:

Date: 5/12/2022

Write the correct answer in the table (30 marks)

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
Q11	**Q12**	**Q13**	**Q14**	**Q15**					

Q1) $\lim_{x\to 0}\frac{\sin 2x^2}{3x^2}=$

a) 1 b) $\frac{2}{3}$ c) 0 d) $\frac{4}{9}$

Q2) If $f(x)=\frac{x+1}{\sqrt{3-|x|}}$, $g(x)=\sqrt{2-x}$, then $D_{f-g}=$

a) $(-3,2]$ b) $[-3,3]/\{2\}$ c) $(-\infty,-3),[2,\infty)$ d) $(-\infty,2]/\{3\}$

Q3) If $f(x)=2\sqrt{4-x^2}-1$, then $R_f=$

a) $[1,6]$ b) $[-1,3]$ c) $[0,2]$ d) $(-\infty,2],[2,\infty)$

Q4) The discontinuity points for $f(x)=\frac{\cot x}{x^2-1}$ are

a) $x=\pm 1,\pm\frac{(2n+1)}{2}\pi$ n=0,1,2,.. b) $x=\pm 1,\pm\frac{n\pi}{2}$ n=0,1,2,.. c) $x=\pm 1,\pm n\pi$ n=0,1,2,.. d) $x=\pm 1,\frac{\pi}{2}\pm n\pi$ n=0,1,2,..

Q5) $\lim_{x\to-\infty}\frac{|x-3|}{3x-2}=$

a) $\frac{1}{3}$ b) $\frac{-1}{3}$ c) 0 d) $-\infty$

Q6) $\lim_{x\to 2^-}\frac{x^2-x-2}{x^2-4x+4}=$

a) $-\infty$ b) $\frac{1}{2}$ c) $+\infty$ d) $\frac{3}{4}$

Q7) $\lim_{x\to 3} \frac{x^2-9}{\sqrt{x^2-6x+9}} =$

a) -6 b) 6 c) -1 d) 1

Q8) $\lim_{x\to+\infty} \frac{\cos x}{2x-3} =$

a) $\frac{1}{2}$ b) $+\infty$ c) 1 d) 0

Q9) If $f(x)=\frac{\sec x}{x+3}$ and $f^{-1}(c)=0$, then c=

a) $\frac{1}{3}$ b) 0 c) $\frac{2}{\sqrt{3}}$ d) $\frac{1}{4}$

Q10) If G is the graph of $f(x)=\frac{1}{x}$, then the graph of $h(x)=\frac{3x+7}{x+2}$ will be

a) Shift G up 7 units and right 2 units b) Shift G down 7 units and left 2 units
c) Shift G down 3 units and left 7 units d) Shift G up 3 units and left 2 units

Q11) One of the following equations is symmetric about origin

a) $y=x\sin x$ b) $y^3=\frac{\cos x}{x}$ c) $5y^3=x+9$ d) $2x^2=3y-1$

Q12) One of the following functions is an even function

a) $f(x)=x\tan^2 x$ b) $f(x)=\frac{\sec 4x}{5x}$ c) $f(x)=\frac{\cos 5x}{2x}$ d) $f(x)=\frac{\sin 2x}{3x}$

Q13) If $f(2x-1)=2-g(3x)$ and $D_f:[-1,1]$, then D_g is

a) $\left[0,\frac{1}{3}\right]$ b) $[-3,3]$ c) $[0,3]$ d) $[1,3]$

Q14) If $f(x)=\frac{4x^2}{ax^2+bx}$, the horizontal and vertical asymptotes respectively are $y=2$, $x=1$, then the values of a and b are

a) a = 2, b=2 b) a = 1, b= −1 c) a = 4, b=8 d) a = 2, b= −2

Q15) If $f(x)=x^2-2x+1$ where $x\geq 1$, then $f^{-1}(x)=$

a) $\sqrt{x-1}+2x$ b) $\sqrt{x}+1$ c) $\frac{\sqrt{x-1}}{2x}$ d) $\frac{\sqrt{x-2}}{x}$